

ERA 5 calibration to Elexon power

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Updates

New

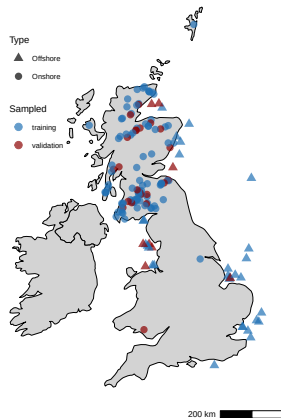
- OOS assessment
 - OOS time
 - OOS locations
- Low wind events distribution

Model validation framework

Framework for OOS assessment

- Run models on a sample of days
 - 80/20 WF split for training and testing
 - Training with 3 days to predict the 4th
 - Randomly chosen 15 days from 2025, 5 per regime
- Evaluate performance at WF level
- Error under aggregation
 - GB level
 - Onshore / offshore

Sampled days for performance assessment

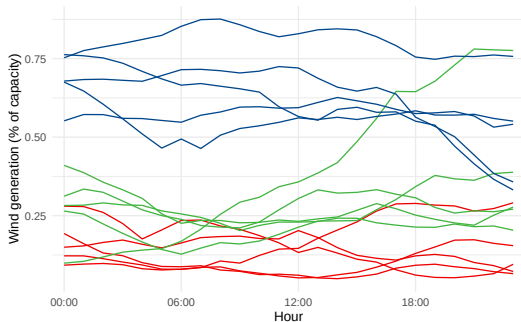


Sampled days in different regimes

- Objective is to assess performance by regime
- OOS days are different regimes of wind speed
- Previous 3 days can be anything

Sampled days for performance assessment

Daily generation time series for sampled days



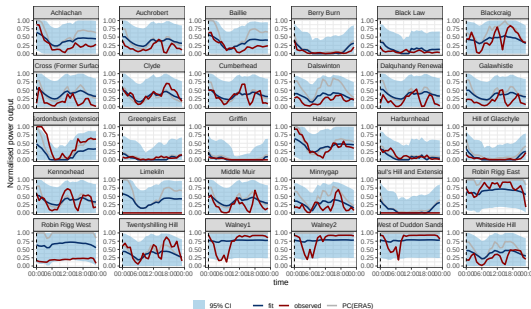
Model portfolio

- Wind farm level models:
 - Generic Power Curve (ERA5 + generic PC estimate)
 - Quantile Mapping
 - Linear model
 - AR1, AR2
 - Spatiotemporal model fine mesh and coarser mesh
- Linear model at Great Britain (GB) level

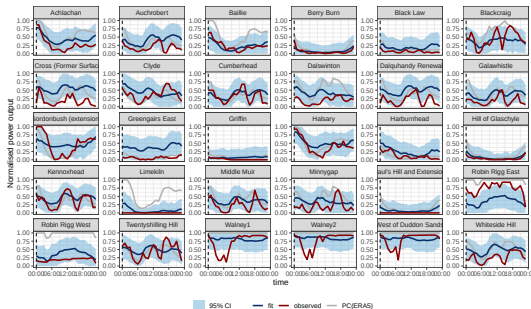
Out-of-sample time

Wind farm prediction bands

Linear model (27-01-2025)

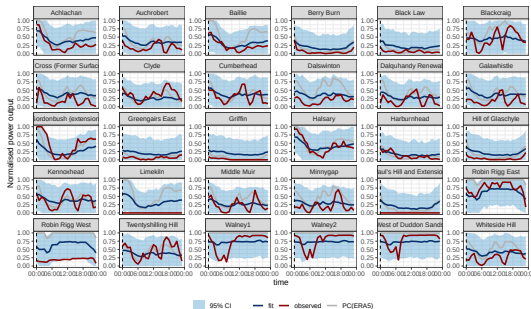


Spatiotemporal model coarse (27-01-2025)

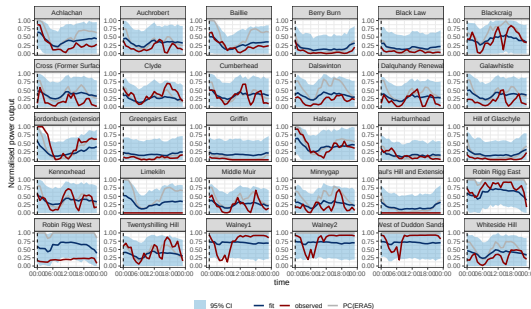


Wind farm prediction bands

1D SPDE model (27-01-2025)

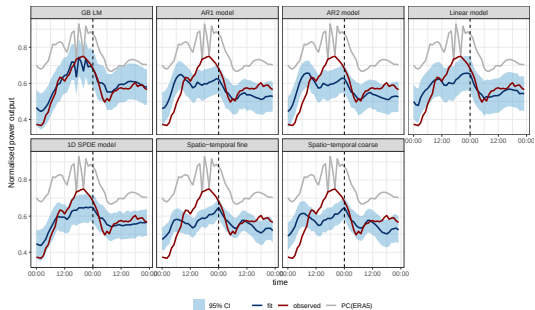


AR1 model (27-01-2025)

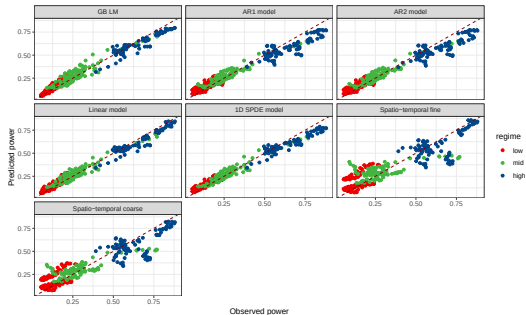


Aggregated prediction bands

GB level



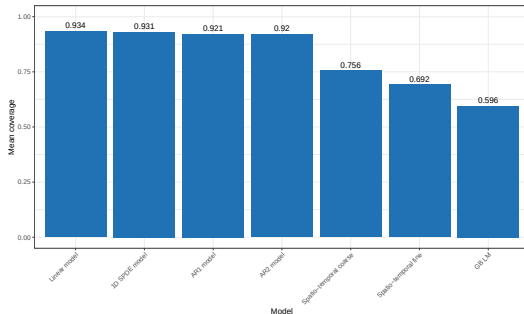
Scatter plot oos



Day-ahead forecast coverage

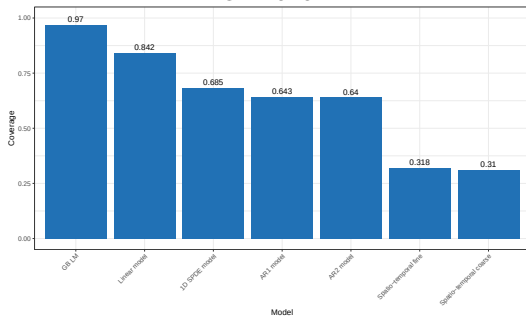
- Models at WF can achieve target coverage
- Aggregation reduces coverage
- High regime more susceptible to coverage loss
- Spatiotemporal model is not improving coverage

Coverage for day-ahead predictions

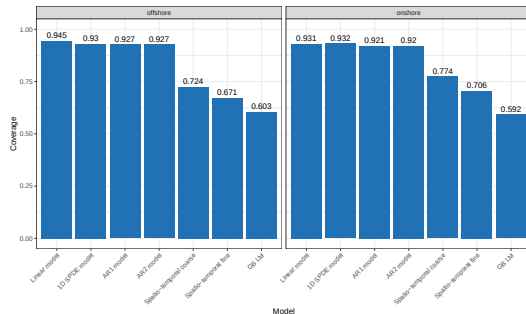


Coverage under aggregation

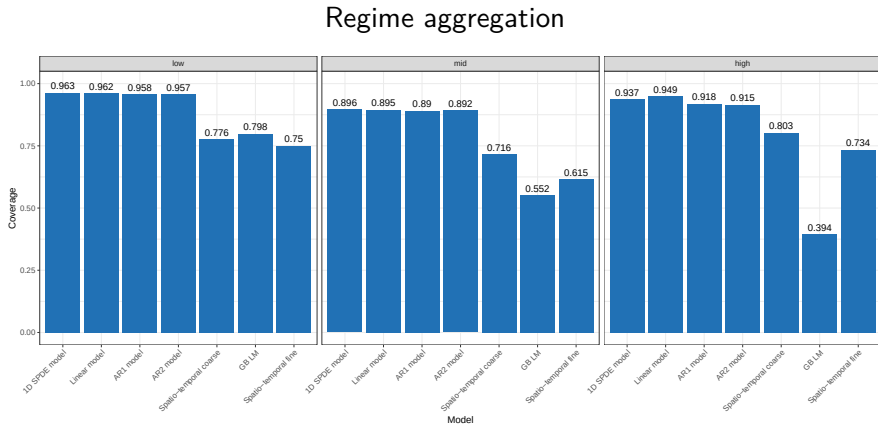
GB level



Onshore / offshore



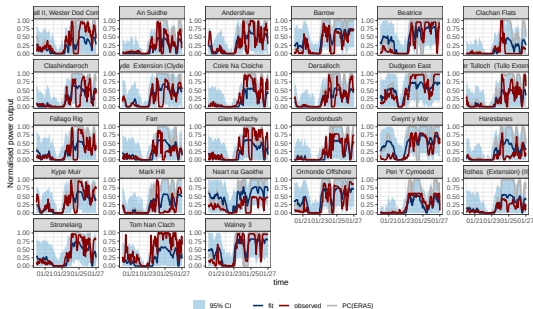
Coverage under aggregation by regime



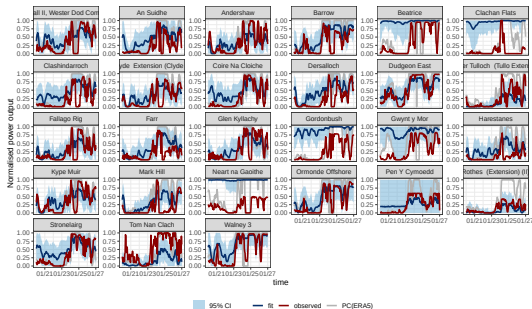
Out-of-sample locations

Wind farm prediction bands

Linear model (27-01-2025)

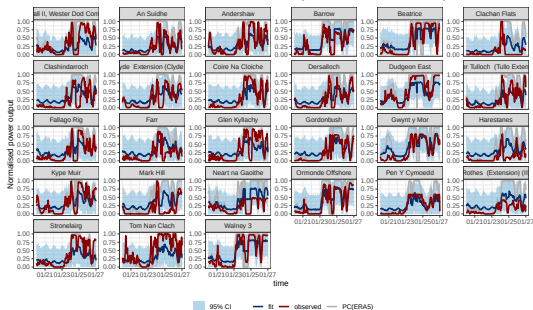


Spatiotemporal model coarse (27-01-2025)

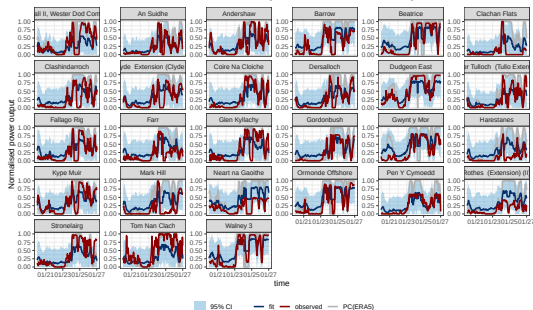


Wind farm prediction bands

1D SPDE model (27-01-2025)

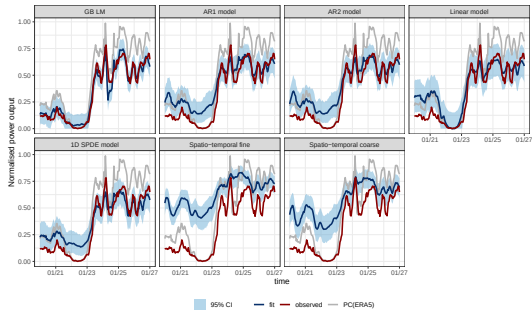


AR1 model (27-01-2025)

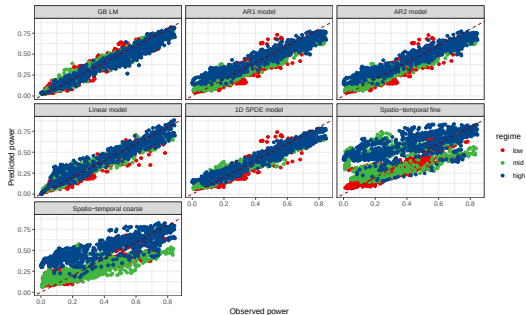


Aggregated prediction bands

GB level (27-01-2025)

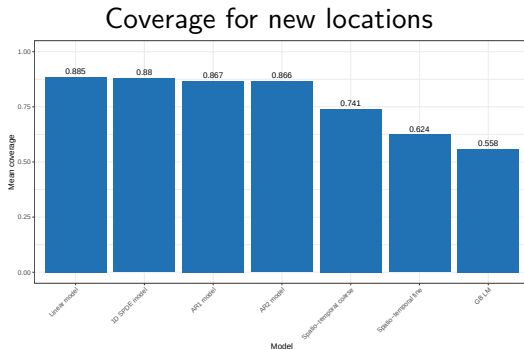


Scatter plot oos



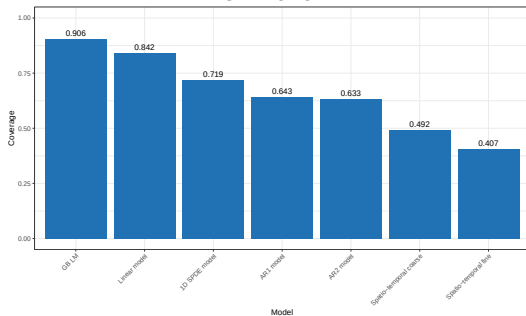
Unseen locations prediction intervals coverage

- Attaining coverage in unseen locations is more challenging
- Isolated locations can lead to high uncertainty
- Location specific events may affect coverage
- Spatiotemporal model is not improving coverage

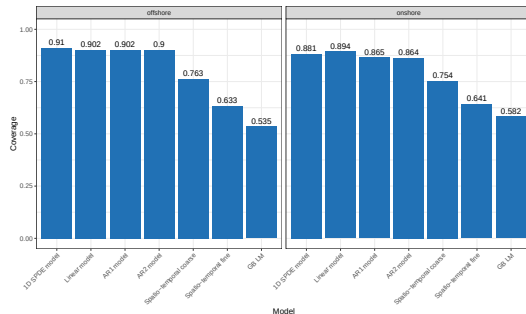


Coverage under aggregation

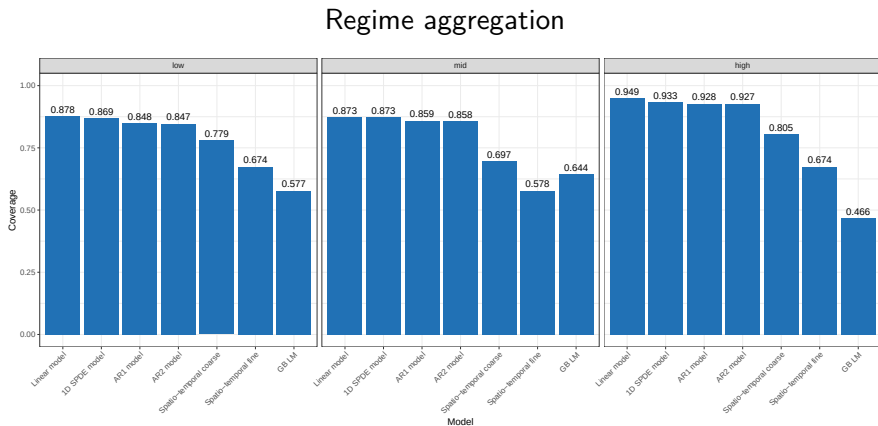
GB level



Onshore / offshore



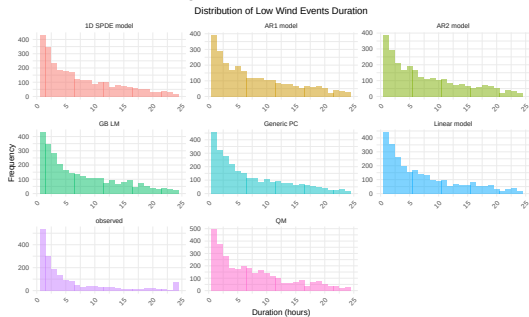
Coverage under aggregation by regime



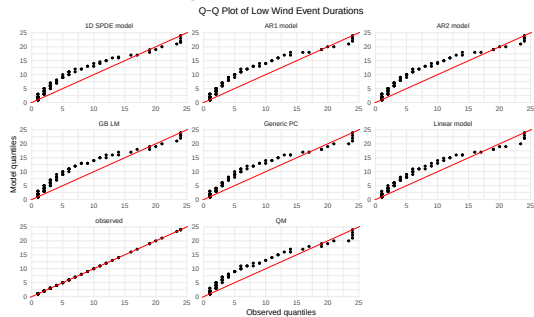
Low wind speed events distribution

Low wind speed events (insample)

Low wind speed events distribution

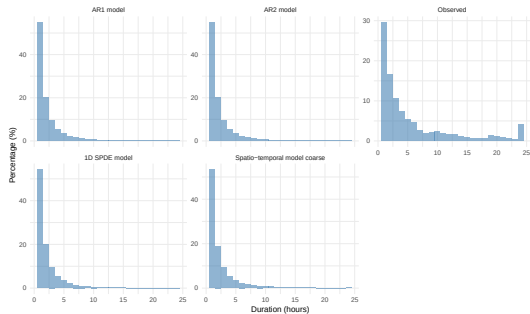


Low wind speed events distribution

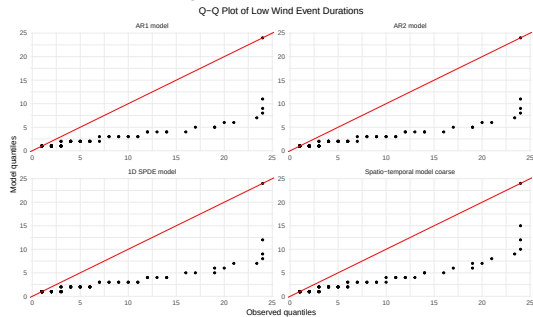


Low wind speed events (oos time)

Low wind speed events distribution

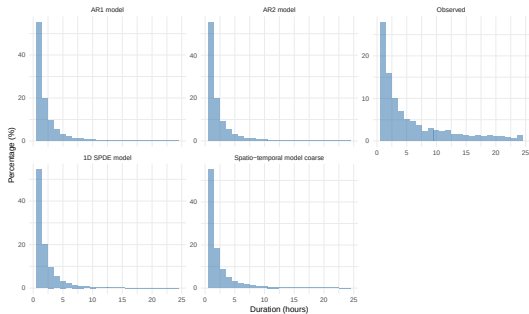


Low wind speed events distribution

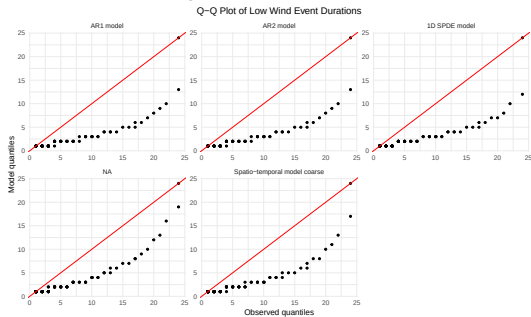


Low wind speed events (oos space)

Low wind speed events distribution



Low wind speed events distribution



Next steps

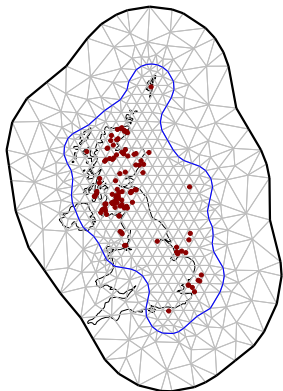
Addressing challenges

- Collapsing uncertainty under aggregation
- Spatiotemporal model is not improving calibration
- Low wind speed events are not well represented in the model samples

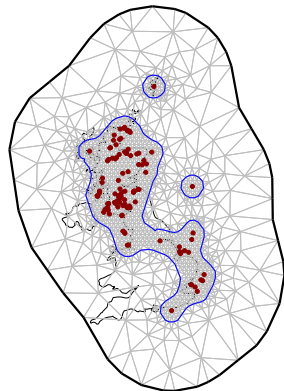
Appendix

Meshes for spatiotemporal models

Coarse mesh



Fine mesh



Error metrics and performance assessment

Error metrics

model	RMSE	MAE	Bias
QM	0.210	0.135	1.58e-04
GB LM	0.191	0.130	2.12e-02
Linear model	0.168	0.117	1.76e-03
1D SPDE model	0.167	0.116	1.46e-03
AR2 model	0.164	0.113	1.45e-03
AR1 model	0.164	0.113	1.44e-03
Spatio-temporal coarse	0.124	0.082	1.26e-03
Spatio-temporal fine	0.097	0.061	1.16e-03

Error metrics by regime

Model	Low			Mid			High		
	RMSE	Bias	MAE	RMSE	Bias	MAE	RMSE	Bias	MAE
QM	0.110	0.062	-1.59e-03	0.196	0.133	1.08e-04	0.305	0.219	2.21e-03
GB LM	0.108	0.069	2.05e-02	0.186	0.132	2.44e-02	0.262	0.193	1.44e-02
1D SPDE model	0.093	0.059	7.34e-03	0.160	0.115	2.50e-03	0.234	0.179	-7.56e-03
Linear model	0.092	0.059	6.57e-03	0.162	0.117	2.26e-03	0.235	0.183	-4.75e-03
AR2 model	0.089	0.056	3.07e-03	0.156	0.112	1.24e-03	0.230	0.176	1.49e-04
AR1 model	0.089	0.056	3.05e-03	0.156	0.112	1.22e-03	0.230	0.176	1.85e-04
Spatio-temporal coarse	0.061	0.037	2.22e-03	0.117	0.081	1.13e-03	0.180	0.132	5.07e-04
Spatio-temporal fine	0.048	0.028	1.75e-03	0.088	0.059	1.05e-03	0.145	0.101	7.84e-04

Error metrics by technology

Model	Offshore			Onshore		
	RMSE	Bias	MAE	RMSE	Bias	MAE
QM	0.213	0.139	5.85e-02	0.209	0.133	-1.85e-02
GB LM	0.184	0.127	8.26e-03	0.193	0.131	2.54e-02
1D SPDE model	0.172	0.119	6.88e-04	0.166	0.114	1.70e-03
AR1 model	0.169	0.117	6.72e-04	0.162	0.111	1.68e-03
AR2 model	0.169	0.117	7.36e-04	0.162	0.111	1.67e-03
Linear model	0.157	0.113	2.84e-03	0.172	0.118	1.42e-03
Spatio-temporal coarse	0.098	0.066	1.26e-03	0.131	0.087	1.26e-03
Spatio-temporal fine	0.073	0.048	5.82e-04	0.103	0.065	1.34e-03

Spatiotemporal calibration model

Model overview

$$\mathbb{E}(p_{it}^{\text{calib}}) = \alpha + \beta_k + \beta_m + \beta_h + s_p(\hat{p}_{it}) + s_w(w_{it}) + u_{it}, \quad (1)$$

- Calibrated power: corrected output at site i , time t .
- α : intercept.
- Fixed effects $\beta_k, \beta_m, \beta_h$: location/season/time structure.
- ERA5 correction s_p : bias adjustment of ERA5 + generic PC estimate.
- Wind effect s_w : nonlinear wind–power correction.
- Spatiotemporal effect u_{it} : Spatial component + AR1 error.

References

- Cannon, D. J., D. J. Brayshaw, J. Methven, P. J. Coker, and D. Lenaghan. 2015. "Using Reanalysis Data to Quantify Extreme Wind Power Generation Statistics: A 33 Year Case Study in Great Britain." *Renewable Energy* 75 (March): 767–78. <https://centaur.reading.ac.uk/38448/>.
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